📖 Official Documentation

3DR Solo Web Utility Owner's Manual

The definitive guide for installing, configuring, updating, and operating the on-board web management application for the 3DR Solo drone.

⚡ Windows Installer (.exe)

🤖 Android Installer (.apk)

📱 Sololink Viewer Android App (.apk)

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1. System Architecture & Overview

The 3DR Solo Web Utility is an open-source, on-board management server and flight deck application designed specifically for the 3DR Solo drone running Open Solo. Rather than relying on cloud servers or legacy mobile applications, the Web Utility operates natively on the Solo's internal companion computer.

> **Key Concept: On-Board Execution**
> The server runs directly inside the Solo's i.MX6 Linux operating system on port `80`, making it instantly accessible from any web browser on any smartphone, tablet, or PC connected to the SoloLink Wi-Fi network.

Network Topology

- **Copter Companion Computer:** `10.1.1.10` (Web Server & Telemetry Host)
- **Artoo Controller:** `10.1.1.1` (Internal SSH Gateway & NAT Router)
- **Pixhawk Autopilot:** Connected to the Solo's computer by serial link (`/dev/ttymxc1` at 921600 baud), shared by Open Solo's telemetry forwarder

2. One-Click Installation & Setup

The Web Utility is installed once, from a Windows PC with `Solo Web Utility v1.2.0 Installer.exe` or from an Android phone with `Solo Web Utility v1.2.0 Installer.apk`. Both are on open-solo.com and install exactly the same files. The Android installer is only an installer: once the utility is on the Solo you can delete it and use a browser.

Step-by-Step Installation Procedure:

1. Power on your 3DR Solo Copter and Artoo Controller.
2. Connect your PC or phone to the Solo's Wi-Fi access point (e.g. `SoloLink_XXXXXX`).
3. Run `Solo Web Utility v1.2.0 Installer.exe` (or install and open the Android installer).
4. The installer connects to the Copter at `10.1.1.10` over SSH, copies the application to `/log/SoloWebUtility2` (on the 6.5 GB `/log` partition, linked as `/usr/bin/SoloWebUtility2`) and the firmware files to `/log/firmware`, and checks every file.
5. The installer adds the web server to `/etc/inittab`, so it starts on every power-up and restarts by itself if it ever stops, and starts it straight away. Its log is `/log/solowebutility.log`.
6. Open `http://10.1.1.10` in a browser on any device connected to SoloLink and log in.

```
# Verification via SSH:
ssh root@10.1.1.10 (Password: TjSDBkAu)
ps | grep solo_web_app.py
# Output: root 1245 /usr/bin/python /usr/bin/SoloWebUtility2/solo_web_app.py
```

> **Already built into the Open Solo image:**
> A **Full Install** (section 3) installs an Open Solo image that already contains the Web Utility, so it survives the update.

3. Automated Firmware Update Engine

The Web Utility ships with ArduCopter 4.7.2 for both the stock Black Cube and the Green Cube, with the Solo's parameter defaults built into the firmware. **Check for Updates** builds a newer

stable ArduCopter on ArduPilot's custom build server when one is available, adds the Solo defaults, and keeps the older versions so you can go back. You can also upload your own `.apj` file.

⬛ Stock Solo Cube (Black)

Flashes `arducopter_CubeSolo_4.7.2.apj`, built for the original Stock Black Cube autopilot.

🟩 Green Cube

Flashes `arducopter_CubeGreen_4.7.2.apj`, built for the upgraded Green Cube.

Flashing Options:

- **Full Install:** Flashes the Pixhawk Cube autopilot AND stages the full Open Solo Linux operating system images (`3dr-solo.tar.gz` and `3dr-controller.tar.gz`).
- **Cube Only:** Flashes only the Pixhawk Cube APJ binary while preserving current Linux OS partitions.

> **About 6 minutes of rebooting:**
> A Full Install reboots the Copter and the Controller to apply their updates. The page shows a countdown that survives page refreshes. The phone or PC loses its connection meanwhile; reconnect to SoloLink when it has finished.

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4. Wi-Fi & Internet NAT Router

The Web Utility features a built-in Wi-Fi manager that connects your Artoo Controller to your home 2.4 GHz Wi-Fi network while maintaining the active SoloLink access point connection.

Bi-Directional IP Forwarding:

When connected to home Wi-Fi, the Controller enables kernel IP forwarding and `iptables` NAT masquerading. This allows your laptop or tablet on your home network to access the Solo (`10.1.1.10`) directly without disconnecting from home internet.

Volatile Credential Safety:
Home Wi-Fi credentials are saved strictly in RAM at `/tmp/wpa_client.conf`. Credential files and non-default routes are automatically cleared upon reboot to guarantee default SoloLink AP recovery.

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5. Telemetry & Cockpit HUD Flight Screen

The Cockpit HUD tab renders an interactive Primary Flight Display (PFD) receiving real-time MAVLink telemetry streams over MAVLink UDP port `14560`.

Live Flight Displays:

- **Attitude Artificial Horizon:** Live pitch and roll synthetic horizon indicator.
- **Speed & Altitude Tapes:** Real-time Airspeed, Groundspeed, Altitude (MSL/AGL), and Compass Heading.
- **System Gauges:** Battery Voltage, Current Draw, Battery Remaining %, GPS Satellites & HDOP, Flight Mode, and Armed/Disarmed Safety Status.
- **Messages Bar:** The Solo's status messages (for example "PreArm: ..." or "EKF ...") appear in a bar over the display, coloured by severity, and dims after a few seconds. Tap or click the bar to see the recent messages.
- **Live Video Stream:** Low-latency H.264 video from the Solo's HDMI input. It recovers by itself when the video stream restarts, and **Restart Video** reconnects it by hand.

One video viewer at a time:
The Controller sends the Solo's video to only one app at a time. While Solex (or another app) shows the video, the Cockpit video stays black, and the other way round. Telemetry, parameters and everything else in the Web Utility work alongside Solex.

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6. Sensor Calibration Wizard

The Web Utility provides a step-by-step post-flash sensor calibration pipeline to ensure your Solo is fully calibrated and flight-ready.

Calibration Sequence:

1. **6-Axis Accelerometer & IMU Calibration:** Samples level, left, right, nose-down, nose-up, and back positions.
2. **Autopilot Reboot:** Automatically sends `MAV_CMD_PREFLIGHT_REBOOT_SHUTDOWN` to commit IMU offsets to persistent EKF storage.
3. **Compass Calibration ("The Solo Dance"):** Dual-magnetometer calibration with live 3-axis rotation cards and real-time fitness scores.

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7. Gimbal Firmware Update

On ArduCopter 4.x the Solo's own boot-time gimbal updater can no longer reach the gimbal. A replacement gimbal can therefore be stuck on old firmware: the GoPro controls are missing, or the gimbal does not follow the tilt paddle. **Update Gimbal Firmware**, at the bottom of the Firmware Update tab, installs the gimbal firmware that ships with Open Solo (`/firmware/gimbal_firmware_1.3.6.ax`).

> **Before you start:**
> Props off and the Solo on the ground. **Do not power off the Solo** until the update has finished (about 5 minutes). The Cube restarts several times and the gimbal does not respond to the tilt paddle meanwhile.

What it does:

1. **Checks:** refuses to start while the vehicle is armed or another update or calibration is running, and remembers the Cube's `MNT1_TYPE`.
2. **Opens a path to the gimbal:** sets `MNT1_TYPE` to 2 and then to 0, rebooting only the Cube each time (the Solo's computer and the web page keep running), and reads the gimbal's firmware version.
3. **Flashes:** if the gimbal is older than the file (or *Reflash* is ticked), uploads the firmware. This takes about 2 minutes, has a time limit, and is retried once.
4. **Puts everything back:** sets `MNT1_TYPE` back and reboots the Cube, even when a step failed.

Afterwards, check that the gimbal levels the camera, follows the tilt paddle, and that the GoPro controls work in your app. The page shows each step and a log; the full log is `/log/gimbal_update.log`.

If something goes wrong:

- **Closing the browser or losing Wi-Fi** does not stop the update: it runs on the Solo. Reopen the page to watch it.
- **Power lost during the update:** the next time the Solo starts, the Web Utility puts `MNT1_TYPE` back by itself. A gimbal left without firmware (LED flashing yellow, camera limp) is recovered by running the update again.

Boot-time gimbal updater:

Open Solo checks the gimbal at every boot. If a gimbal flash is ever interrupted, the original updater retries it with no time limit and can hold up the Solo's startup (no SSH, no pairing) until it is power-cycled. The card's **Install time-limited updater** button replaces it with a version that has time limits and logs to `/log/gimbal_update.log`; **Restore original updater** puts the original back. The replacement has not yet been tested on a drone.

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8. ArduPilot Parameters & Calibrated Presets

Manage, search, edit, and save over 1,000 ArduPilot flight parameters with instant MAVLink ACK verification.

One-Click Hardware Presets:

- **Compass:** Here 1 / Here 2 GPS compass, or the stock Solo leg compass.
- **Retractable Landing Gear** and **Bait Drop / Payload Release** servo setups.
- **Flight Performance:** *Fast* (quicker, crisper handling, 40° lean, 8 m/s in Fly) or *Default* (the Solo's stock feel; undoes Fast).
- **Battery:** failsafe thresholds for the stock 4S LiPo, for 4S Li-ion high-capacity packs, or for 5S Li-ion packs on a BMSOne.

> **Need new batteries?**
> Original Solo batteries were last made in 2015. The **Battery Solutions** tab compares the ready-built Battery Mod, the do-it-yourself *Solo Battery Mod for Dummies* guide (included on the Solo as a PDF) and the BMSOne, and applies the right preset for your pack.

9. Artoo Controller Button Customization

The Controller Setup tab presents an interactive SVG diagram of the 3DR Artoo Controller for customizing physical hardware button mappings.

Supported Buttons:

Configure Short Hold and Long Hold actions for: **A Button, B Button, Preset 1, Preset 2, and Camera Button**. Mappings are written directly to `/usr/bin/extSettings.conf` on the Solo.

10. Sololink Viewer Android APK

For mobile pilots who prefer a dedicated Android app for FPV video and On-Screen Display (OSD), the Sololink Viewer APK is available for direct installation. In the Web Utility, use 📥 **Viewer for Android** in the top bar to download it.

Installation Instructions:

1. Download `sololink_viewer.apk` directly onto your Android device.
2. Enable "Install from Unknown Sources" in Android Security settings.
3. Open the APK file to install the application.
4. Connect your Android device to the SoloLink Wi-Fi network and launch Sololink Viewer for low-latency RTSP video.

11. Safety Interlocks & Security

> **Armed Safety Interlock:**
> Firmware flashing, hostapd cycling, Wi-Fi credential modifications, and autopilot reboots are hard-locked whenever `cockpit_mgr.is_vehicle_armed()` returns `True`. Arming the vehicle automatically disconnects home Wi-Fi mode to ensure 100% telemetry link priority.

Authentication & Password Management:

- Default password: `sololink`. While it is still in use, the utility asks you to change it after logging in; **Keep Default** skips this for the rest of the browser session.
- Change the password with the 🔒 button at the top right. This signs out every other device.
- Each login gets its own random session that expires after 24 hours without use.
- After 5 wrong passwords within 5 minutes, logins from that device are paused for 30 seconds, doubling up to 5 minutes for repeated attempts.
- Passwords are hashed using SHA-256 with random salt strings and stored in `auth.json`.
- CLI password reset script: `python change_password.py --reset`

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12. Official Downloads & Community Links

⚡ Solo Web Utility v1.2.0 Installer.exe (Windows)

🤖 Solo Web Utility v1.2.0 Installer.apk (Android)

📱 sololink_viewer.apk (Android App)

3DR Solo Web Utility · Built for the Open Solo Community.
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